

# Dissolving the Three Great Problems of Cognitive Architecture

A scale-invariant, buildable account of binding, the hard problem, and the one-and-the-many

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## Abstract

Three problems have guarded the philosophy of mind and the theory of cognitive architecture for generations: the *binding problem* (how distributed processing becomes one experience), the *hard problem* (why physical process is accompanied by felt experience), and the *one-and-the-many* (how an integrated system coordinates globally without flattening its specialized parts). We argue that each has been protected less by genuine difficulty than by a shared frame error: the search for a *continuous, permanent* locus of unity — a central theater, an enduring substrate, a global field. Replacing that with a *transactional* account — unity as an episodic, query-triggered state-space contraction over otherwise-parallel streams — dissolves all three. The account is not merely philosophical: it is an engineering specification, it carries falsifiable predictions, and its central quantity (the *content-capacity residue*) is computable. We exhibit a small computation showing that this residue is a well-defined entanglement monotone, independent of the separate “magic” resource it is easily confused with. The distinguishing claim of this paper is constructive: these are not dissolutions to be admired but architectures to be built and tested.

## 1 Introduction

A problem can be hard for two reasons. It can be hard because the world is genuinely intricate, or it can be hard because we have framed it so that no answer of the kind we are looking for could exist. The three classic problems of mind and cognitive architecture, we will argue, are mostly of the second kind. Each is generated by an implicit demand for a *standing* thing: a place where the many streams of processing are *always already* unified (binding); a property that *continuously* accompanies the physical (the hard problem); a global state that is *permanently* maintained (the one-and-the-many). Drop the demand for permanence, and each problem reshapes into something with a clean, mechanical answer.

The reframe is a single idea applied three times. A mind — biological or artificial — is, at baseline, a population of *decentralized, parallel, structurally fragmented* processors. It does not hold a continuous global state. Instead, unity is *instantiated on demand*: a top-down query or an environmental exigency triggers a rapid contraction of the system’s state space into a single, synchronized frame, which then relaxes back into parallel divergence. Coherence is a transaction, not a status. We call this stance *transactionalism*, and develop it for binding (Sec. 2), the hard problem (Sec. 3), and the one-and-the-many (Sec. 4).

What separates this account from a century of dissolutions is that it is *buildable*. Section 4 is an engineering specification, not a metaphor; Section 5 states predictions that can fail; and the quantity at the center of Section 3 — the residue that we identify with felt texture — is one we can compute and exhibit (Fig. 2). A philosopher dissolves a problem with an argument. We aim to

dissolve these by *instantiating the architecture in which they do not arise* and showing it makes testable claims.

**A minimal apparatus.** We need only two commitments. First, the unit of analysis is an *informational stream*: a localized processor with a state, a set of accessible configurations, and a tendency that biases how it moves through them. Second, any such stream admits two modes of description that do not reduce to one another: an *extrinsic* structural mode  $F_1$  (the stream tracked from outside as routing, dynamics, mechanism) and an *intrinsic* mode  $F_2$  (the stream grasped from its own vantage).  $F_1$  and  $F_2$  are *parallel projections* of one underlying process; neither is constructible from the other by any operation living wholly within the other’s description. This single non-factoring is the seed of the hard problem, and naming it precisely is what lets us dissolve rather than re-pose it.

## 2 The Binding Problem: Episodic Functionalism

The binding problem asks how spatially and functionally distributed processing — color here, motion there, meaning elsewhere — becomes one unified experience. The traditional search is for the place where it all comes together: a Cartesian theater, a global workspace held continuously lit, an enduring substrate of consciousness. That search is the frame error. There is no such standing place, and there does not need to be.

**Metastable modularity.** Under baseline conditions, the system’s processors run concurrently as decentralized, parallel, structurally fragmented streams. Crucially, the architecture *tolerates divergence*: when immediate integration is not required, the streams may hold mutually incommensurable partial states — a controlled superposition — at no cost. Coherence is not a field that must be continuously sustained against entropy; it is a capacity that is exercised when needed.

**Transactional integration.** Synchronic phenomenal unity is instantiated *dynamically*, as an episodic transaction. A salience or exigency trigger — the arrival of a payload that some stream or the environment makes urgent — precipitates a non-linear phase transition: a rapid contraction of the joint state space (Fig. 1). The decentralized streams undergo vector-space convergence and broadcast a single, synchronized frame. We deliberately avoid calling this trigger “top-down”: there is no central executive that *issues* it (Sec. 4 makes the mechanism explicit). “Top-down” would name only the *direction of the resulting convergence* (many streams into one frame), never a controller giving orders — a distinction we return to below.

**The horizon of convergence.** At the moment of the transaction, two things happen that a dualist about appearance-versus-mechanism would keep separate: the *appearance* of an integrated perspective, and the *objective functional convergence* of the underlying systems. We claim these are *token-identical* — the same event, described two ways (the  $F_1/F_2$  split of Sec. 3), not two correlated events. Unity is not discovered to be present; it is *enacted*, and only at the boundary of action. Between transactions there is no unified subject to be missing — which is why, from the inside, the gaps are not experienced as gaps.

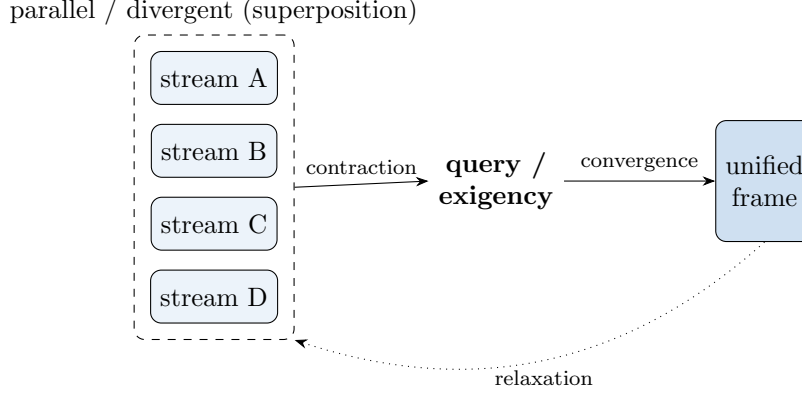


Figure 1: **Binding as transaction.** Baseline: streams run in parallel and may diverge (superposition, left). A query or exigency triggers a state-space contraction (center) that converges the streams into one synchronized frame (right); the system then relaxes back to parallel. Unity is the contraction event, not a maintained field.

### 3 The Hard Problem: Perspectival Physicalism

Levine’s explanatory gap survives by forcing a choice: either strip experience of its qualitative interiority (reductive materialism), or posit a new, non-physical law of nature (property dualism). Both moves accept the framing that experience is a *further fact* requiring derivation from, or addition to, the physical. We reject the framing.

**The geometrical contraction.** The transaction of Sec. 2 is, described structurally, the contraction of a high-dimensional informational manifold into a single, localized, egocentric coordinate system — many degrees of freedom gathered into one point of view. This is an ontologically singular event.

**Asymmetric modes of presentation.** That one event has two aspects, fixed entirely by mode of presentation. Tracked from *without*, in the structural mode  $F_1$ , it is deterministic routing and convergence — a functional description. Grasped from *within*, in the intrinsic mode  $F_2$ , it is a subject-centered phenomenal perspective. There is no gap to bridge between them because there are not two events: there is one event with two irreducible projections. The “explanatory gap” is the shadow cast by demanding that  $F_2$  be *derived* from  $F_1$ , when the two are parallel, not serial. This is a *dissolution*, not a *solution*: we do not explain how routing produces experience; we show that the demand to do so mistakes one event for a chain of two.

**Subjective texture as structural residue.** If experience is the intrinsic aspect of the contraction, what fixes its specific texture? Not an ignition at a complexity threshold. The texture is the *content-capacity residue* of the transaction: the difference between a stream taken in isolation and that same stream operating inside its wider context. Formally, for a stream  $S$  embedded in a whole, let  $\eta(S)$  measure how much the whole reshapes the part. A stream that is a clean, separable factor of its context ( $\eta = 0$ ) has no residue — nothing of the context is written into it. A stream deeply interwoven with its context carries a large residue: the context is present in the part as the very texture of how the part is.

**The residue is computable — and is not what it is often confused with.** This is the paper’s empirical anchor. Model a stream as a subsystem of a joint quantum state; let  $\eta(S) = 1 - \text{Tr } \rho_S^2$  be the mixedness of its reduced state (zero iff  $S$  is a clean factor; maximal iff  $S$  is maximally interwoven with its context). One might suspect this residue is the same as computational *magic* (non-stabilizerness) [19, 20] — the resource that, in recent holographic models [21, 22, 23], distinguishes inert geometry from dynamical geometry. It is not. Figure 2 shows a double dissociation: the residue  $\eta$  tracks *entanglement* (part–whole correlation) exactly, and is *completely independent of magic*. A maximally interwoven but stabilizer (zero-magic) state has  $\eta = 0.5$ ; a high-magic but separable state has  $\eta = 0$ . The felt residue is the *correlation between part and whole* — not the separate generative resource it superficially resembles. We flag this because conflating the two is an easy and tempting error.<sup>1</sup>

**The residue is substrate-general; the quantum model is an instrument.** We use a quantum density-matrix model because it makes the residue cleanly computable and lets us draw the magic dissociation sharply — *not* because the architecture requires quantum hardware (it does not; see Sec. 4). What the dissolution actually rests on is part–whole *non-factorability*: the irreducibility of a part’s in-context state to its in-isolation state. That is substrate-general. In the quantum model it is entanglement ( $\eta = 1 - \text{Tr } \rho_S^2$ ); in a classical distributed architecture it is the *mutual information* between a node and its context — the gap between the node’s conditional behavior given its context and its marginal behavior in isolation. These are the same structural quantity. The source of the classical mixedness is not an observer’s ignorance but the *architecture itself*: in a decentralized, concurrent system the global joint state may be fully deterministic while each node’s *observed marginal* behaves as a mixed state, because asynchronous execution and partition prevent the part from being read off a single fixed snapshot of the whole. The residue is thus not only computable in the quantum model but directly *measurable* in the running classical system (Sec. 5). One may object that classical mixedness is “merely epistemic” (uncertainty over well-defined states) whereas quantum mixedness is “ontological” (non-separability). But that distinction is itself drawn from the extrinsic  $F_1$  stance; from the intrinsic  $F_2$  vantage there is no “mere” uncertainty, only the contextual shaping that constitutes the texture. The dissolution turns on non-factorability, which is real in both regimes; the epistemic/ontological gloss is one more instance of the  $F_1/F_2$  asymmetry this section describes. The residue is thus not a quantum artifact but a substrate-invariant: computed here in the cleanest available model.

## 4 The One-and-the-Many: Scale-Invariant Transactionalism

The one-and-the-many asks how an integrated architecture achieves global coordination (the One) without flattening the specialized local diversity that makes its parts competent (the Many). Here the transactional stance becomes a concrete design.

**Rejection of the monolith.** The usual routes to global harmony are a single massive model, or a shared universal interlingua into which every domain translates. Both destroy the Many. A universal interlingua in particular is an epistemic trap: forcing physics, law, and biology through one common vocabulary flattens exactly the domain-specific syntax that expertise consists in.

**Zero-degree-of-freedom binding.** Instead, global unity is maintained as a *structural invariant* — a binding with zero degrees of freedom (Fig. 3). Partition the system into a stratified hierarchy of

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<sup>1</sup>We made it ourselves and caught it by computing Fig. 2; the figure is the correction.

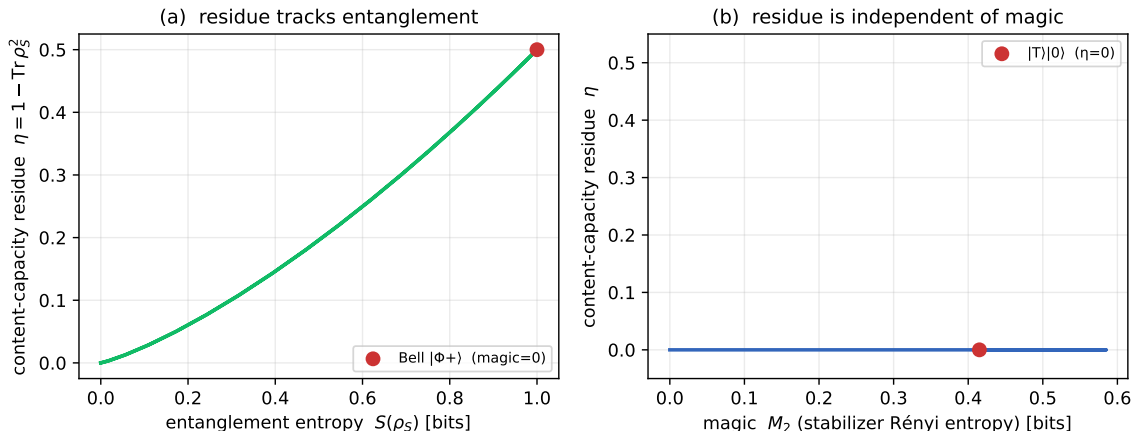


Figure 2: **The content-capacity residue is a computable entanglement monotone, independent of magic.** (a) Across a family of two-qubit states, the residue  $\eta = 1 - \text{Tr} \rho_S^2$  rises monotonically with the entanglement entropy of the part; the Bell state (red) realizes the maximum at zero magic. (b) Across a family with the part held separable, magic ( $M_2$ , stabilizer Rényi entropy) ranges widely while  $\eta$  stays identically zero; the  $|T\rangle|0\rangle$  state (red) has substantial magic and zero residue. The residue is part-whole correlation, full stop.

*namespaced* domain-expert nodes (in the spirit of, but distinct from, sparsely-gated mixtures of experts [24]). Route messages *deterministically by payload type*, not through a learned (and therefore biased, and unauditable) mediator. Because routing is type-matching, the binding layer carries no parameters to train: it cannot drift, cannot acquire a preference, cannot become a hidden monolith. Cross-domain communication is pushed to *explicit, point-to-point typed bridges* at the domain seams, each an auditable translation between two local vocabularies — never a flattening into one.

**On scaling, and the interlingua it must not become.** Naively, point-to-point bridges across  $N$  domains threaten an  $O(N^2)$  explosion, and the obvious escape — abstracting payload types until they cover arbitrary crossings — would silently reconstruct the very universal interlingua we reject. We avoid both. Bridges are built *lazily*: one is instantiated only when a crossing is actually demanded, and indexed in a registry that is a switchboard of specific transforms, not a semantic node of its own. The realized bridge count therefore tracks *actual* cross-domain demand — empirically sparse — not  $N^2$ . Crucially, a bridge is a specific  $A::x \rightarrow B::y$  transform between two *namespaced, domain-local* types; it never generalizes into a shared schema, so the type system cannot creep into an interlingua. We state this as a design discipline, not a solved theorem: the precise upper bound on typological complexity — how expressive payload types may grow before locality erodes — is an open question we flag rather than claim to have closed.

**The lazy synthesized cache.** Global consensus is never enforced as a permanent baseline. The collective state is a *lazy synthesized cache*: on a query it crystallizes *locally and on-demand* into a unified macro-state by collapsing exactly the relevant nodes; between queries the components drift safely back into parallel superposition; and the cache self-heals at the next relevant execution. This is the binding transaction of Sec. 2 at the architectural scale — which is why we call the account *scale-invariant*: the same mechanism (parallel-until-measured, unified-on-demand) resolves binding within a mind and coordination across a society of experts. Coherence is transactional,

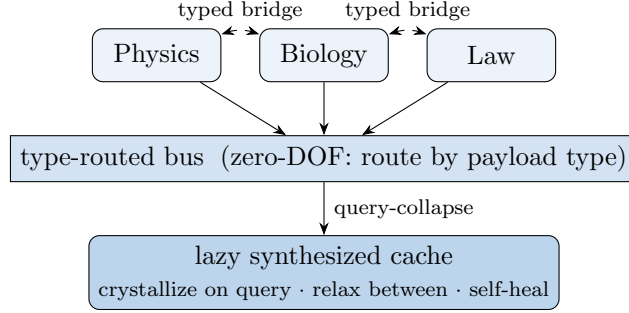


Figure 3: **The One held without flattening the Many.** Namespaced domain-expert nodes (top) keep their local syntax. The binding bus routes deterministically by payload type and so carries zero trainable degrees of freedom (it cannot drift into a biased monolith). Genuine cross-domain transfer goes through explicit typed bridges at the seams. Global consensus is a lazy cache that crystallizes on-query and relaxes between queries.

demand-driven, and intrinsically self-balancing: you pay for the coherence you use, when you use it, and not before.

**No homunculus initiates the transaction.** This architecture is what dissolves the residual worry left by Sec. 2: if unity is *triggered*, who triggers it? The answer is that nobody does. The trigger is the *arrival of a typed payload* — emitted by a node or supplied by the environment — and the contraction is scoped by the zero-DOF type-router, which has no parameters, no preferences, and no vantage from which to “decide.” There is no one who asks; there is a payload that arrives and a deterministic routing that collapses exactly the nodes whose types match. This is precisely why the account stays faithful to the no-central-self tradition it inherits: the appearance of a coordinating “I” at the moment of convergence is the convergence *viewed from within* ( $F_2$ ), not a controller standing behind it. “Top-down” (Sec. 2) thus names the geometry of the event — many gathered into one — and never smuggles in an executive.

## 5 Falsifiable Predictions

Because the account is an architecture, it can be wrong in specific, checkable ways. Among its predictions:

1. **Separation.** Sub-systems whose objectives operate on *separate* degrees of freedom outperform otherwise-matched systems whose objectives share degrees of freedom (destructive interference). Falsified if shared-DOF systems perform equally.
2. **On-demand convergence.** Integrated, “bound” representations are present transiently around task/query events and absent between them, rather than maintained continuously. Falsified if a continuous global binding signal is found that is necessary for function.
3. **Per-domain memory.** A society of specialized nodes with local, per-domain memory outperforms a monolithic shared memory of equal capacity — with the gain non-monotonic in node strength (mid-capacity nodes benefit most). (*Independently consistent with recent results on task-focused memorization in multimodal agents [25].*)

4. **No interlingua advantage.** Replacing typed point-to-point bridges with a universal interlingua degrades cross-domain task fidelity, the loss growing with domain heterogeneity.
5. **Residue tracks correlation, not generation.** The structural correlate of an integration event’s “richness” (the content-capacity residue) tracks part-whole correlation, not the separate non-stabilizer/generative resource (Fig. 2).

**Measuring the residue in a running system.** Prediction 5 is testable in a classical implementation by *ablation*, requiring no quantum hardware. For a node  $S$ , run it (i) embedded in the live system and (ii) isolated from the bus, and compare its output distributions over a matched input set; the divergence between the in-context and in-isolation distributions — the mutual information  $I(S; \text{context})$  — is exactly  $\eta(S)$ . A clean, separable node yields  $\eta \approx 0$ ; a node whose behavior is substantially rewritten by its context yields a large residue. This is the standard lesion methodology, repurposed as a phenomenology probe: the felt-texture correlate becomes an ordinary, repeatable measurement on a deployed architecture.

**A note on long-term coherence.** If binding is transient, what carries macro-narrative consistency across episodes? Not a continuously maintained global thread — there is none. Consistency is re-synthesized: the history lives *fragmented* across local, namespaced node memories (Prediction 3), and each episodic contraction re-assembles a coherent macro-state from those shards on demand. Long-term identity is therefore a recurring structural *resonance*, not a stored chronicle — which is why it can survive interruptions, partitions, and the loss of any single shard.

## 6 Related Work

The binding problem was sharpened by feature-integration theory [8] and posed as a modeling challenge by von der Malsburg [9]. Global Workspace theory [10, 11] and Integrated Information Theory [12] both seek the locus of integration; we agree integration is real but deny it is *continuously maintained* — binding is the workspace transiently lighting up, not its standing illumination. Our refusal of a permanent locus of unity continues the anti-Cartesian-theater tradition of Dennett [6] and Metzinger’s self-model theory [7], while supplying the positive mechanism (the on-demand transaction) those critiques leave implicit.

The hard problem [3, 4] and the explanatory gap [2], with Nagel’s what-it-is-like [1] and Block’s access/phenomenal distinction [5], set the terms we dissolve. Our  $F_1/F_2$  dual-aspect is a contemporary, computationally grounded form of the neutral-monist lineage — Mach [13], James [14], Russell [15], Whitehead [16], and Strawson’s realistic monism [17] — differing in that we identify the dual-aspected event concretely (the manifold contraction) and make its residue *computable* (Fig. 2).

The residue computation rests on standard quantum-information notions: the Gottesman–Knill theorem [18], magic states [19], and the stabilizer Rényi entropy [20] as a magic monotone. The contrast it draws on — entanglement structures geometry, a further resource grants dynamics — parallels the holographic programme of Ryu–Takayanagi [21] and Van Raamsdonk [22], with the recent magic-gives-gravity extension [23]. The architecture of Sec. 4 draws on sparsely-gated mixtures of experts [24] but replaces the *learned* gate with deterministic type-routing; its per-domain-memory prediction is independently supported by recent task-focused memorization results [25].

## 7 Discussion

The account is deliberately deflationary about *permanence* and inflationary about *events*. It keeps the phenomena every theory must keep — unity is real, experience is real, expertise is real — while denying the standing entities (the theater, the extra property, the monolith) that made them seem mysterious. Against global-workspace theories, binding is the workspace’s *lighting up*, not its continuous illumination. Against property dualism and reductive materialism alike, the hard problem dissolves because  $F_1$  and  $F_2$  are parallel projections of one event, so neither the elimination of  $F_2$  nor a new law connecting it to  $F_1$  is required. Against monolithic integration, the One is a zero-DOF invariant plus a lazy cache, not a flattening.

We stress what is and is not claimed for Sec. 3. We do not derive experience from structure; that would re-pose the gap. We claim the demand for such a derivation mistakes a single two-aspected event for a causal chain. Whether one finds this satisfying is partly temperamental — but the account earns its keep elsewhere, in Sec. 4’s buildable architecture and Sec. 5’s falsifiable claims, which a purely verbal dissolution cannot offer.

## 8 Conclusion

Three problems, one move: replace the standing locus of unity with the on-demand transaction. Binding becomes the convergence event; the hard problem becomes the two aspects of that event, with felt texture the computable residue of part-whole correlation; the one-and-the-many becomes a zero-DOF binding over namespaced experts with a lazy synthesized cache. The wager of this paper is that these are not positions to be defended but systems to be built — and that a mind so built will exhibit the predictions of Sec. 5. The mysticism was vestigial. What remains is architecture.

## A The residue computation

Figure 2 is generated by `figures/make_eta_figure.py`. For a pure two-qubit state  $|\psi\rangle$  with subsystem  $S$ ,  $\eta(S) = 1 - \text{Tr} \rho_S^2$  where  $\rho_S = \text{Tr}_E |\psi\rangle\langle\psi|$ . Magic is the stabilizer 2-Rényi entropy  $M_2 = -\log_2(d \sum_P \Xi_P^2)$  with  $\Xi_P = \langle\psi|P|\psi\rangle^2/d$  over all Pauli strings  $P$  and  $d = 2^n$ . The double dissociation (Bell:  $\eta = 0.5$ ,  $M_2 = 0$ ;  $|T\rangle|0\rangle$ :  $\eta = 0$ ,  $M_2 = 0.415$ ) establishes that  $\eta$  is an entanglement monotone independent of magic.

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